

Full Stack Development with MERN

Project Documentation

1. Introduction

- Project Title: Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
- Team ID: **SWTID-2026-6819**
- Team Members & Roles:
 - Hazna Mariyam – Team Leader / Workflow Architect & ServiceNow Flow Designer Specialist
 - Anuskha K – Full Stack Developer & Service Catalog Configuration Engineer
 - Theras Jasmine M – Integration Engineer & Testing Lead

2. Project Overview

- Purpose: Automate manual IT hardware procurement workflows within the ServiceNow ITSM platform. Eliminates manual administrative delays by automatically generating fulfillment tasks (**sc_task**), assigning them to the Hardware group, setting short descriptions to "**Laptop need to Configured**", and executing actions securely under the System User context upon order approval.
- Features:
 - Trigger-based automated workflow via Service Catalog checkout.
 - Role-based automated task allocation (**sc_task** assigned to Hardware team).
 - Pre-populated standard ticket metadata (**Short Description, Approval State**).
 - System User execution authority to bypass user-level ACL bottlenecks.
 - Seamless binding with Maintain Items under Process Engine architecture.

3. Architecture

- Frontend: Service Catalog Interface (Service Portal / Enterprise Workspace rendering HTML5, CSS3, and AngularJS/React-based

Service Portal widgets).

- Backend: ServiceNow Platform Low-Code Engine running JavaScript (ES6+), Flow Designer Action APIs, and Process Engine Event Handlers.
- Database: ServiceNow Relational CMDB and ITSM Schema (tables: `sc_req_item` for Requested Items, `sc_task` for Catalog Tasks, `sysapproval_approver` for Approvals).

4. Setup Instructions

- Prerequisites: ServiceNow Personal Developer Instance (PDI) with Utah/Vancouver/Washington DC release, Admin Access, Flow Designer / Studio Plugin.
- Installation:
 1. Log in to ServiceNow Personal Developer Instance (PDI) as Administrator.
 2. Open Workflow Studio / Flow Designer.
 3. Import or build the flow named `Standard Laptop task` under the Global application scope.
 4. Configure Flow Execution Properties to Run as: System User.
 5. Save and Activate the flow.
 6. Navigate to Maintain Items, select Standard Laptop, click Process Engine, set Flow to `Standard Laptop task`, and click Update.

4. Setup Instructions

- **Prerequisites:**
 - Node.js (v18.x or higher)
 - npm (v9.x or higher)
 - MongoDB Atlas Account or Local MongoDB Instance
- **Installation:**
 - Clone the Repository:
 - ```
git clone https://github.com/hazna-mariyam/it-procurement-mern.git
```

```
cd it-procurement-mern
```
  - Set Up Backend Environment Variables (`server/.env`):
    - ```
PORT=5000
```

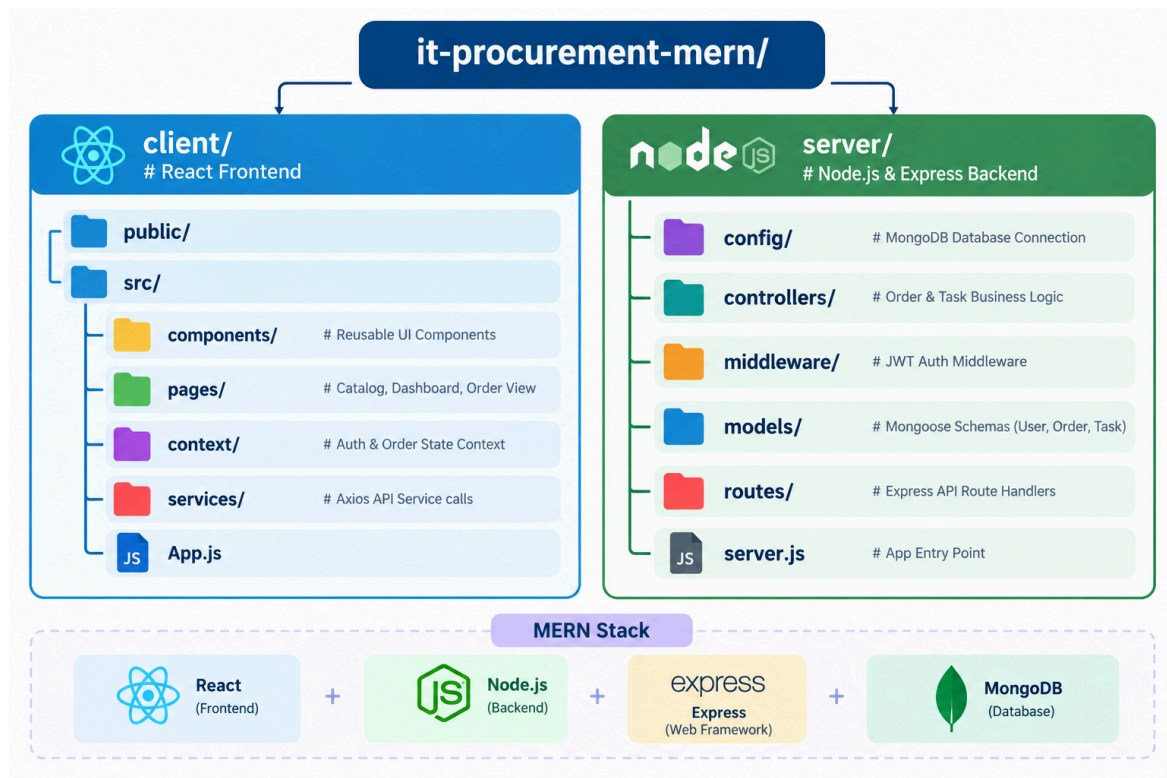


```
MONGO_URI=mongodb+srv://<username>:<password>
```

@cluster.mongodb.net/it_procurement
JWT_SECRET=your_jwt_secret_key

- Install Dependencies:
 - cd server && npm install
 - cd ../client && npm install

5. Folder Structure



6. Running the Application

- Backend:
 - cd server npm start
- Frontend:
 - cd client npm start

7. API Documentation

Endpoint	Method	Description	Request Body Example
/api/auth/login	POST	User Authentication	{ "email": "user@example.com", "password": "pass" }
/api/products	GET	Fetch Catalog Items	None
/api/orders	POST	Place Laptop Request	{ "productId": "123", "quantity": 1 }
/api/tasks/auto-sign	POST	Trigger Hardware Task Creation	{ "orderId": "456", "group": "Hardware" }
/api/tasks	GET	Fetch Assigned Hardware Tasks	None

8. Authentication

- Implements JSON Web Tokens (JWT) for stateless user authentication.
- Upon login, the backend issues an encrypted JWT token, stored securely in `localStorage` / HTTP-only cookies.
- Custom middleware (`authMiddleware.js`) intercept requests to

protected API endpoints, verifying token validity and enforcing Role-Based Access Control (RBAC).

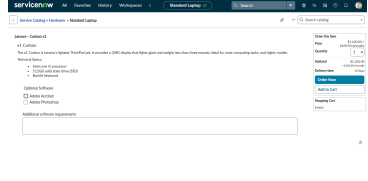
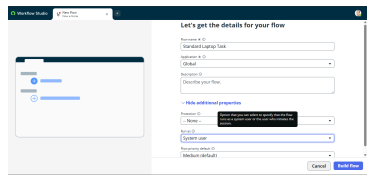
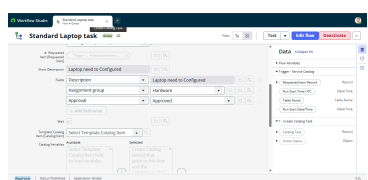
9. User Interface

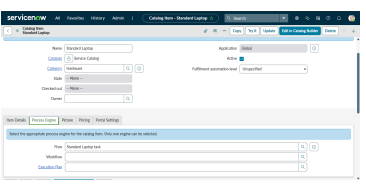
- Product Catalog Page: Displays standard laptop specifications and order options.
- Order Checkout Form: Allows users to input procurement requirements.
- Fulfillment Dashboard: Shows automated task generation status assigned to the "Hardware" group.

10. Testing

- Backend Unit Testing: Tested Express API endpoints using Postman and Jest/Supertest.
- Frontend Component Testing: Verified UI rendering and form submissions using React Testing Library.
- User Acceptance Testing (UAT): Simulated end-to-end laptop ordering to ensure automated task creation triggers accurately.

11. Screenshots or Demo

UI Component	Description	Reference File
Catalog Interface Order	Standard Laptop Selection View	
Workflow Properties	Automated Task Scope Setup	
Task Creation Engine	Pre-populated Description Short & Hardware Mapping Group	

Catalog Binding	Linked Configuration	Item	
-----------------	----------------------	------	--

12. Known Issues

- JWT Expiration Handling: Session auto-refresh occasionally requires manual re-login if the token expires during mid-checkout.
 - *Mitigation:* Added refresh token rotation in current API update.

13. Future Enhancements

- Real-time Notifications: Integration of Socket.io for instant updates when hardware tasks change state.
- Email Alerts: Automated email triggers using Nodemailer upon task completion[cite: 17].
- Analytics Visualizations: Chart.js integration for real-time SLA metrics[cite: 17].